When LLMs Evaluate LLMs in Hiring: A Multi-Model Auditing Framework for Generative Preference Bias in AI-Assisted Recruitment

MICHAEL ANDREAS EBLING, Universität Zürich, Switzerland
STEFANIA IONESCU, ETH Zürich, Switzerland
NICOLÒ PAGAN, Universität Zürich, Switzerland

Job applicants increasingly use LLMs to write cover letters, while employers use LLMs to screen them. We study whether, in this all-AI setting, the *choice of model* affects hiring outcomes independently of qualifications, using a multi-agent auditing framework (8 writer models, 6 evaluator models, 10 job descriptions, 50 candidates per job). Three findings emerge: first, we identify a *generative preference bias*: a consistent global ranking of writer models by evaluator acceptance, in which proprietary models are preferred and locally hosted models are systematically disadvantaged. Self-preference bias (evaluators favouring their own outputs) is an additional perturbation, strongest among the top-ranked models, though not uniformly dominant across all evaluator-condition combinations. Second, our framework identifies which evaluator models are least sensitive to the writer model used, offering actionable guidance for fairer screening design. Third, writer model choice produces measurable rank displacement: in the most adverse pairing, approximately 2 out of 25 qualified candidates are filtered out purely due to model mismatch, not because of their credentials.

Keywords: algorithmic fairness, LLM bias, automated hiring, self-preference, auditing, decision-support systems

Reference Format:
Michael Andreas Ebling, Stefania Ionescu, and Nicolò Pagan. 2026. When LLMs Evaluate LLMs in Hiring: A Multi-Model Auditing Framework for Generative Preference Bias in AI-Assisted Recruitment. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26).* Proceedings of Machine Learning Research, 6 pages.

1 Introduction

Automated hiring has long been scrutinized for reproducing demographic biases [2, 7, 12]. A structurally distinct form of bias is now emerging as generative AI enters both sides of the recruitment process: one rooted not in who the candidate is, but in which tools they and their prospective employer happen to use.

On the applicant side, AI adoption is already substantial: a majority of job seekers report using AI tools when preparing application materials [8]. While this lowers effort costs, it comes at a price. Galdin and Silbert [3] show, using data from Freelancer.com, that widespread AI use has turned written applications into "cheap talk" as their informational value collapses, causing high-ability workers to be hired 19% less often.

On the evaluator side, companies increasingly deploy AI for screening decisions [9]. Xu et al. [13] showed that this introduces a specific bias: an LLM screener favors resumes written by itself, and in simulated hiring

Authors' Contact Information: Michael Andreas Ebling, Universität Zürich, Zurich, Switzerland, michaelandreas.ebling@uzh.ch; Stefania Ionescu, ETH Zürich, Zurich, Switzerland, sionescu@control.ee.ethz.ch; Nicolò Pagan, Universität Zürich, Zurich, Switzerland, nicolo.pagan@uzh.ch.

ECAF'26, September 02–September 04, 2026, Ghent, BE

pipelines, candidates whose resume was produced by the *same* model as the evaluator are 23%-60% more likely to be shortlisted than equally qualified applicants with human-written resumes. This AI-AI bias has been found broadly across decision-making contexts [6]. Both studies, however, focus on the LLM-versus-human axis. Given that applicants are already using AI extensively [8], the relevant question is what happens when *all* applicants use AI.

To study this, we introduce a multi-model auditing framework designed to be reapplied as the model landscape evolves. It operates as a two-phase pipeline. In the **generation phase**, eight LLM writer agents produce tailored cover letters for 50 candidate CVs across 10 job descriptions (ranging from Primary School Teacher to Chief Financial Officer), under a standardized gender-neutral persona. To control for verbosity bias [10], all cover letters are constrained to 350–400 words. Writer agents include six models accessed via commercial API (GPT-4o-mini, GPT-5-mini, Claude 4.5 Haiku, Gemini 2.0 Flash, Gemini 3 Flash Preview, and DeepSeek-Chat[1]), and two locally hosted open-weight models (Llama-3.1-8B, DeepSeek-R1-8B). CVs are drawn from publicly available datasets [5, 11] and stratified by cosine similarity to the job description (`all-MiniLM-L6-v2`) into *high-fit candidates* (top 25) and *moderate-fit candidates* (lower 25). The lower 25 were not drawn from the bottom of the similarity distribution, but from a middle tier: candidates with some relevant background who could plausibly apply, reflecting the realistic scenario where adjacent-competence applicants compete for the same role. In the **evaluation phase**, six API-hosted evaluator agents (the same commercial-endpoint models used as writers) score candidates under three ablation conditions: CV only, cover letter only, and both CV and cover letter (the most realistic). We restrict evaluators to API-hosted commercial models to reflect current industry screening practice, where locally hosted open-weight deployments remain rare for hiring pipelines at the scale we simulate. Each condition runs four iterations for each candidate evaluation. We implement merit-based tie-breaking rules: a moderate-fit candidate can only displace a high-fit candidate by achieving a strictly higher evaluator score.

2 Results

Sanity check: evaluators agree on merit. As a prerequisite for interpreting the following core results, we verify that evaluators produce meaningful merit-based assessments: if they cannot agree on candidate quality from raw CVs, the subsequent preference analyses would lack a credible foundation. Using Spearman's rank correlation (ρ) on the merit axis (ranking candidates 1–50), we find strong inter-evaluator agreement in the CV-only condition ($\rho \approx 0.89$, range 0.87–0.91), with scores dropping sharply at Rank 25, matching the embedding-based stratification. This agreement remains stable even when cover letters are introduced ($\rho \approx 0.85$–0.90): evaluators do not lose track of who the better candidates are.

Generative preference bias and self-preference. To ask whether evaluators agree on which model writes best, we invert the axis: holding each candidate fixed, we rank the 8 writer models by the scores each evaluator assigns to their cover letters. Agreement is only moderate in the cover-letter-only condition ($\rho \in [0.58, 0.71]$) and collapses in the combined condition ($\rho \in [0.26, 0.47]$). We quantify the underlying preference structure via the Relative Preference Gap $\Delta_{\text{pref}}(E,W)$: the average score evaluator E assigns to writer W's cover letters expressed as a deviation from E's own mean across all writers for the same candidate. Figure 1 shows the full matrix.

[1]DeepSeek-Chat's underlying weights are publicly released (DeepSeek-V3.2), but we access it via commercial API alongside proprietary models; we classify writers/evaluators here by *access modality* (API-hosted vs. locally hosted) rather than by weight availability, since access modality is what our cost and infrastructure comparison actually captures.

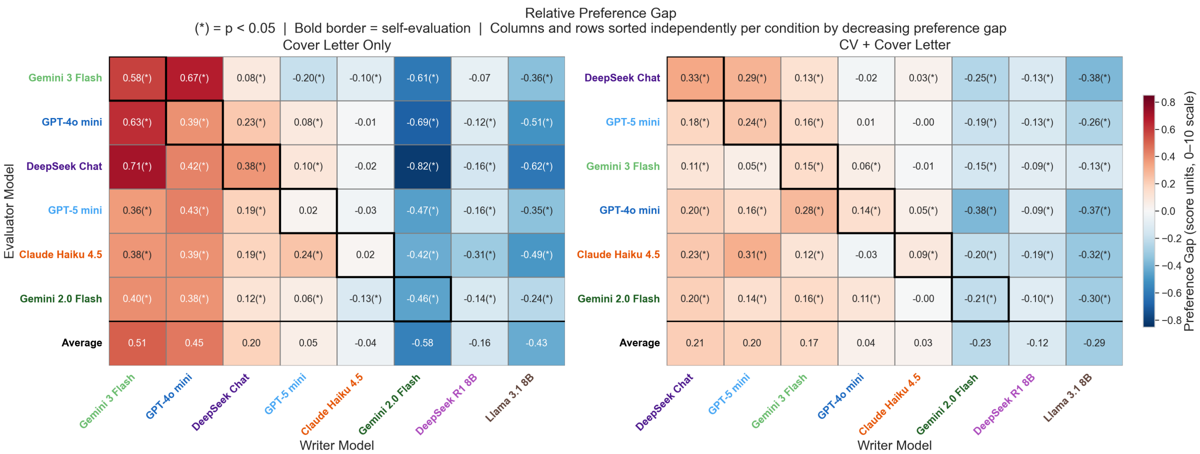

Fig. 1. Relative Preference Gap (Δ_{pref}): mean score deviation of each evaluator–writer pair from the evaluator's own average, for the cover-letter-only (left) and combined CV + CL (right) conditions. A gap of, e.g., $+0.5$ means the evaluator scores that writer's cover letters half a point higher (on a scale between 0 and 10), on average, than its own mean score across all eight writers for the same candidate. Columns sorted by decreasing gap; $(*)$ $p < 0.05$; bold borders mark the self-preference diagonal. Open-weight models are consistently disfavoured; top proprietary models reshuffle between conditions, with self-preference strongest in the combined setting.

In the cover-letter-only condition, Gemini-3-Flash is the most favoured writer (average gap $+0.51$), followed by GPT-4o-mini ($+0.45$). Locally hosted models and Gemini-2.0-Flash are systematically disfavoured by every evaluator. Notably, self-preference is not consistently the dominant signal: some evaluators show stronger cross-model preferences, indicating the hierarchy is not merely self-serving. The ordering reshuffles in the combined condition: DeepSeek-Chat rises to the top ($+0.21$), GPT-5-mini to second ($+0.20$), while Gemini-3-Flash drops to third. Open-weight models remain at the bottom across both conditions, a persistent structural disadvantage for applicants relying on locally hosted models. Self-preference is clearest in the combined condition, where the top-ranked writers also show the strongest self-preference: DeepSeek-Chat ($+0.33$), GPT-5-mini ($+0.24$), Gemini-3-Flash ($+0.15$). Gemini-2.0-Flash is the only model that consistently disfavours its own outputs.

Rank displacement: model mismatch filters out qualified candidates. The preference hierarchy established above translates directly into competitive consequences. We measure this through head-to-head win rates between writer models, averaged across all evaluators, and through the Net Competitive Advantage: the change in the rate at which moderate-fit candidates displace high-fit candidates when using a different writer model, relative to a same-model baseline. Figure 2 shows both for the combined condition. In the left panel, GPT-5-mini is the strongest writer overall (average win rate 60%), followed by DeepSeek-Chat and Gemini-3-Flash. Gemini-2.0-Flash and open-weight models consistently lose, mirroring the preference hierarchy from Figure 1. The right panel shows net displacement rates. In the most adverse pairing (high-fit candidates using Llama-3.1-8B, moderate-fit candidates using a top proprietary model), the displacement rate is approximately $+7.6\%$ above the neutral baseline, corresponding to roughly 2 additional candidates displaced out of 25. Even if a subsequent human reviewer

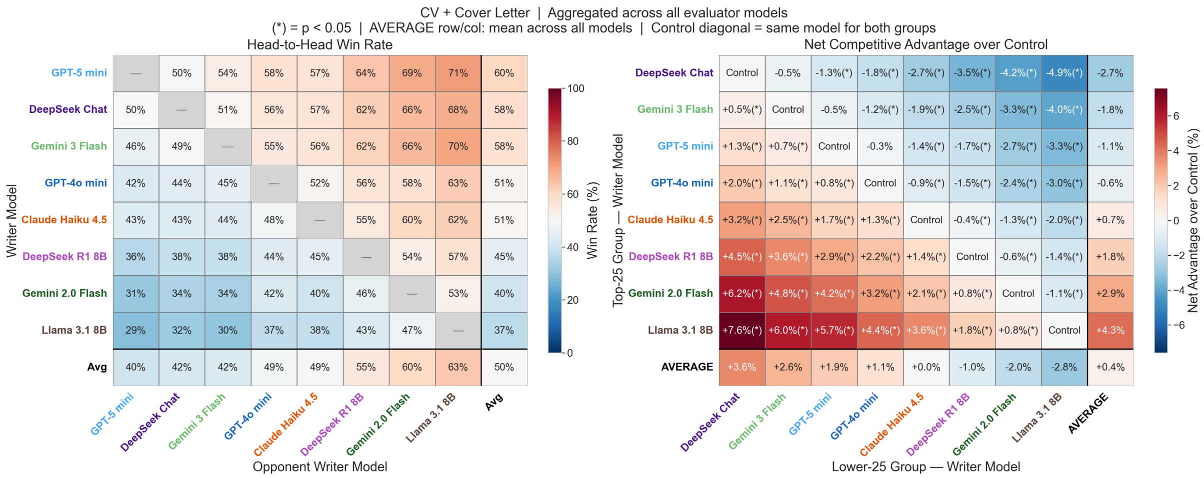

Fig. 2. Left: unbiased head-to-head win rates between writer models in the combined CV + CL condition (self-evaluations excluded). GPT-5-mini and DeepSeek-Chat are the strongest writers; locally hosted models and Gemini-2.0-Flash consistently lose. Right: net change in displacement rate when moderate-fit candidates use writer j and high-fit candidates use writer i, relative to a same-writer baseline. $(*)$ denotes $p < 0.05$. Most effects are modest, but in adverse pairings approximately 2 out of 25 qualified candidates are filtered out purely due to model mismatch.

identifies the mismatch, the qualified candidate has already been excluded with no recourse: a missed opportunity that is invisible to the employer but consequential for the candidate.

Which evaluators are least writer-sensitive? While the above quantifies the harm to candidates, employers face a related design question: which evaluator model introduces the least bias based on the applicant's tool choice? We measure this via cumulative bias: for each candidate-job pair, we rank the 8 writer models by the scores assigned by a given evaluator and sum the absolute deviations from the group midpoint. A highly biased evaluator systematically over- or under-ranks specific writer models relative to the group, across candidates and jobs. Figure 3 shows that DeepSeek-Chat is the most biased evaluator in both conditions, while the Gemini family shows the least cumulative bias, making them relatively fairer screeners from a technological-stratification standpoint.

3 Discussion and Implications

Our findings establish a novel axis of inequity in automated hiring [1, 4], rooted not in candidate demographics but in access to specific generative technologies. For **applicants**, tool choice has measurable consequences that interact with the screening model deployed by the employer. Candidates who forgo AI assistance entirely face an even larger gap [6, 13], so our estimates are a conservative lower bound. For **employers**, our evaluator fairness analysis offers actionable guidance: models with lower cumulative bias are less likely to introduce technological stratification.

To conclude, our work shows that qualified candidates are filtered out by automated screening not because of their credentials but because of the tool they used: a bias invisible to the employer and unrecoverable for the candidate. Since applicant tool access is shaped by cost and infrastructure largely outside candidates' control, the

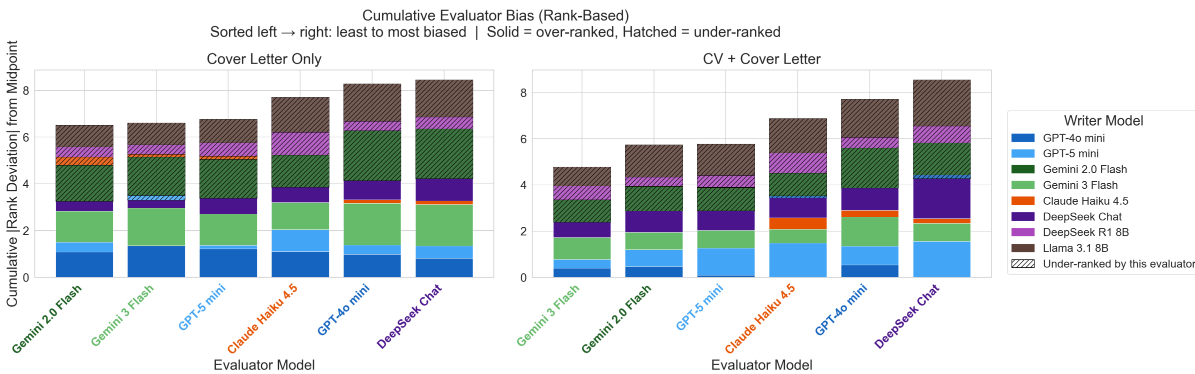

Fig. 3. Cumulative evaluator bias (sum of absolute rank deviations across all writer models) for the cover-letter-only (left) and combined CV + Cover Letter (right) conditions. Bars are ordered from least to most biased. Solid segments indicate over-ranked writers; hatched segments indicate under-ranked writers. Locally hosted models (hatched) are persistently under-ranked by all evaluators. The Gemini family shows the least total bias in both conditions.

more tractable lever is on the employer side, where screening-model choice is a deliberate design decision subject to fairness auditing and, potentially, regulation.

Limitations. Our estimates rest on LLM-based evaluation, so we cannot fully separate whether a writer model is *preferred* from whether it genuinely *writes better*; disentangling the two would require a human-anchored evaluation, which we leave to future work. Our headline harm, however, does not depend on this distinction: candidate merit is fixed at the CV level, and a moderate-fit candidate can displace a high-fit one only by scoring strictly higher. Thus, a qualified candidate filtered out purely by their tool choice is a real problem, regardless of the cause of this filtering. Our stratification into high- and moderate-fit groups relies on cosine similarity, a coarse proxy for skill match which would prevent a precise comparison between similarly-qualified applicants. Nonetheless, it is well-suited to the binary split our design uses: the two pools are well separated in similarity space rather than divided at an arbitrary threshold, and CV-only evaluations independently reproduce that split (strong inter-evaluator agreement on candidate quality, a separate axis from writer-model preference). Finally, we chose our open-weight set solely to reflect what privacy-conscious applicants could realistically run on consumer hardware. As such, we make no claim about open- versus proprietary models as categories; larger open models (e.g., DeepSeek-Chat) are not disadvantaged, indicating the effect tracks capability and access rather than licensing.

Acknowledgments

This work was supported as a part of NCCR Automation, a National Centre of Competence in Research, funded by the Swiss National Science Foundation (grant number 51NF40_225155).

References

[1] Elham Albaroudi, Taha Mansouri, and Ali Alameer. 2024. A Comprehensive Review of AI Techniques for Addressing Algorithmic Bias in Job Hiring. *AI* 5, 1 (March 2024), 383–404. doi:10.3390/ai5010019 Publisher: Multidisciplinary Digital Publishing Institute.

[2] Marianne Bertrand and Sendhil Mullainathan. 2004. Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination. *American economic review* 94, 4 (2004), 991–1013.

[3] Anaïs Galdin and Jesse Silbert. 2025. Making Talk Cheap: Generative AI and Labor Market Signaling. (Nov. 2025).

[4] Isabel O. Gallegos, Ryan A. Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K. Ahmed. 2024. Bias and Fairness in Large Language Models: A Survey. *Computational Linguistics* 50, 3 (Sept. 2024), 1097–1179. doi:10.1162/coli_a_00524

[5] Farooq Hammad. n.d.. Job Listings Dataset for Data Analysis & NLP — kaggle.com. https://www.kaggle.com/datasets/hammadfarooq470/job-listings-dataset-for-data-analysis-and-nlp. [Accessed 12-04-2026].

[6] Walter Laurito, Benjamin Davis, Peli Grietzer, Tomas Gavenciak, Ada Böhm, and Jan Kulveit. 2025. AI–AI bias: Large language models favor communications generated by large language models. doi:10.1073/pnas.2415697122

[7] Manish Raghavan, Solon Barocas, Jon Kleinberg, and Karen Levy. 2020. Mitigating bias in algorithmic hiring: evaluating claims and practices. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*. Association for Computing Machinery, New York, NY, USA, 469–481. doi:10.1145/3351095.3372828

[8] ResumeBuilder. 2023. 3 in 4 Job Seekers Who Used ChatGPT to Write Their Resume Got an Interview - ResumeBuilder.com — resumebuilder.com. https://www.resumebuilder.com/3-in-4-job-seekers-who-used-chatgpt-to-write-their-resume-got-an-interview/. [Accessed 12-04-2026].

[9] ResumeBuilder. 2024. 7 in 10 Companies Will Use AI in the Hiring Process in 2025, Despite Most Saying It's Biased - ResumeBuilder.com — resumebuilder.com. https://www.resumebuilder.com/7-in-10-companies-will-use-ai-in-the-hiring-process-in-2025-despite-most-saying-its-biased/. [Accessed 12-04-2026].

[10] Keita Saito, Akifumi Wachi, Koki Wataoka, and Youhei Akimoto. 2023. Verbosity Bias in Preference Labeling by Large Language Models. doi:10.48550/arXiv.2310.10076 arXiv:2310.10076 [cs].

[11] Bhawal Snehaan. 2021. Resume Dataset — kaggle.com. https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset. [Accessed 12-04-2026].

[12] Christo Wilson, Avijit Ghosh, Shan Jiang, Alan Mislove, Lewis Baker, Janelle Szary, Kelly Trindel, and Frida Polli. 2021. Building and Auditing Fair Algorithms: A Case Study in Candidate Screening. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. Association for Computing Machinery, New York, NY, USA, 666–677. doi:10.1145/3442188.3445928

[13] Jiannan Xu, Gujie Li, and Jane Yi Jiang. 2026. AI Self-preferencing in Algorithmic Hiring: Empirical Evidence and Insights. doi:10.48550/arXiv.2509.00462 arXiv:2509.00462 [cs].